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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

I Semester B. E. Examinations Nov/Dec-19

Common to All Branches

ELEMENTS OF MECHANICAL ENGINEERING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.
3. Use of steam tables is permitted.

PART-A

1	1.1	State first law of thermodynamics.	01
	1.2	Type of steam used for power generation.	01
	1.3	Write the expression for internal latent heat of the steam.	01
	1.4	Pelton turbine is used when the available head is _____.	01
	1.5	_____ turbine is also called as propeller turbine.	01
	1.6	_____ engine develops power for alternate revolution of the crank shaft.	01
	1.7	The type of refrigerator which contains heater cum separator is _____.	01
	1.8	The engine injects fuel at the end of compression stroke is _____.	01
	1.9	The type of operation carried out to produce conical surface is _____.	01
	1.10	The process of finishing the previously drilled hole is known as _____.	01
	1.11	The machine tool used to produce shaft is _____.	01
	1.12	Write any four types of milling operations.	02
	1.13	The type of milling machine used to produce gears is _____.	01
	1.14	Write any two functions of the flux in arc welding process.	02
	1.15	_____ type of flame is produced for equal quantities of oxygen and acetylene gas.	01
	1.16	List the functions of a tailstock of a lathe.	02
	1.17	The type of gear drive used for large reduction in velocity ratio is _____.	01

PART-B

2	a	Discuss the concepts of work & heat in a system.	06
	b	With the help of $T-H$ diagram, describe the formation of steam using steam generating equipment.	06
	c	Define the following properties of steam. i) Enthalpy of wet steam ii) External work of evaporation.	04
3	a	List the classification of water turbines and explain with neat sketch, the working of Kaplan turbine.	08
	b	With the help of suitable diagram, explain the working of vapour compression refrigeration system.	08
		OR	

4	a	Compare open cycle gas turbine & closed cycle gas turbine.	04
	b	Define the following: i) Ton of refrigeration ii) COP of refrigerator.	04
	c	With the help of suitable diagram, explain the working of Pelton wheel	08
5	a	With the help of PV diagram, explain the working of four stroke diesel engine.	08
	b	Draw & label the various parts of IC engine & describe the function of the following components. i) piston and piston rings, ii) connecting rod, iii) crank, iv) cylinder head.	08
		OR	
6	a	Identify the engine which develops the power for every revolution of the crank shaft & works on otto cycle. Describe the working of the engine with suitable diagram.	08
	b	The following readings were taken during the test of a single cylinder, four stroke engine: Cylinder diameter $25cm$, stroke length $40cm$, mean effective pressure $7bar$, engine speed $250\,revolution\,per\,min$, net load on brake drum $1080N$, effective diameter of the brake $1.5m$, fuel used/hour $10kg$, heat value of the fuel $44300kJ/kg$ Calculate the following parameters: i) IP ii) BP iii) Mechanical efficiency iv) indicated thermal efficiency.	08
7	a	With a neat sketch explain any four drilling operations.	08
	b	Explain the following operations: i) Plain milling ii) Angular milling iii) Gang milling iv) Straddle milling.	08
8	a	With the help of suitable diagram, explain the principle of arc welding process.	06
	b	With reference to belt drive system, describe the following concepts: i) creep ii) slip iii) velocity ratio.	06
	c	List the advantages & limitations of CNC machines over conventional machines.	04